



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Stephanos Mermingis
August 26, 2005





Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion



Executive Summary

Summary of methodologies

- Data collection using webscraping and API
- Data Wrangling
- Exploratory Data Analysis
- Interactive Plotly dashboard App
- Using Folium to view launch sites
- Predictive Analysis for launch success



Introduction

- SpaceX launches Falcon 9 rockets and saves money on launches by retrieving the bottom part of the rocket, making them competitive in the space exploration market.
- Using the data from SpaceX, we can predict if it is lucrative to use them to launch our own company's rockets.
- We will predict the probability of successful launches in the future

Section 1

Methodology

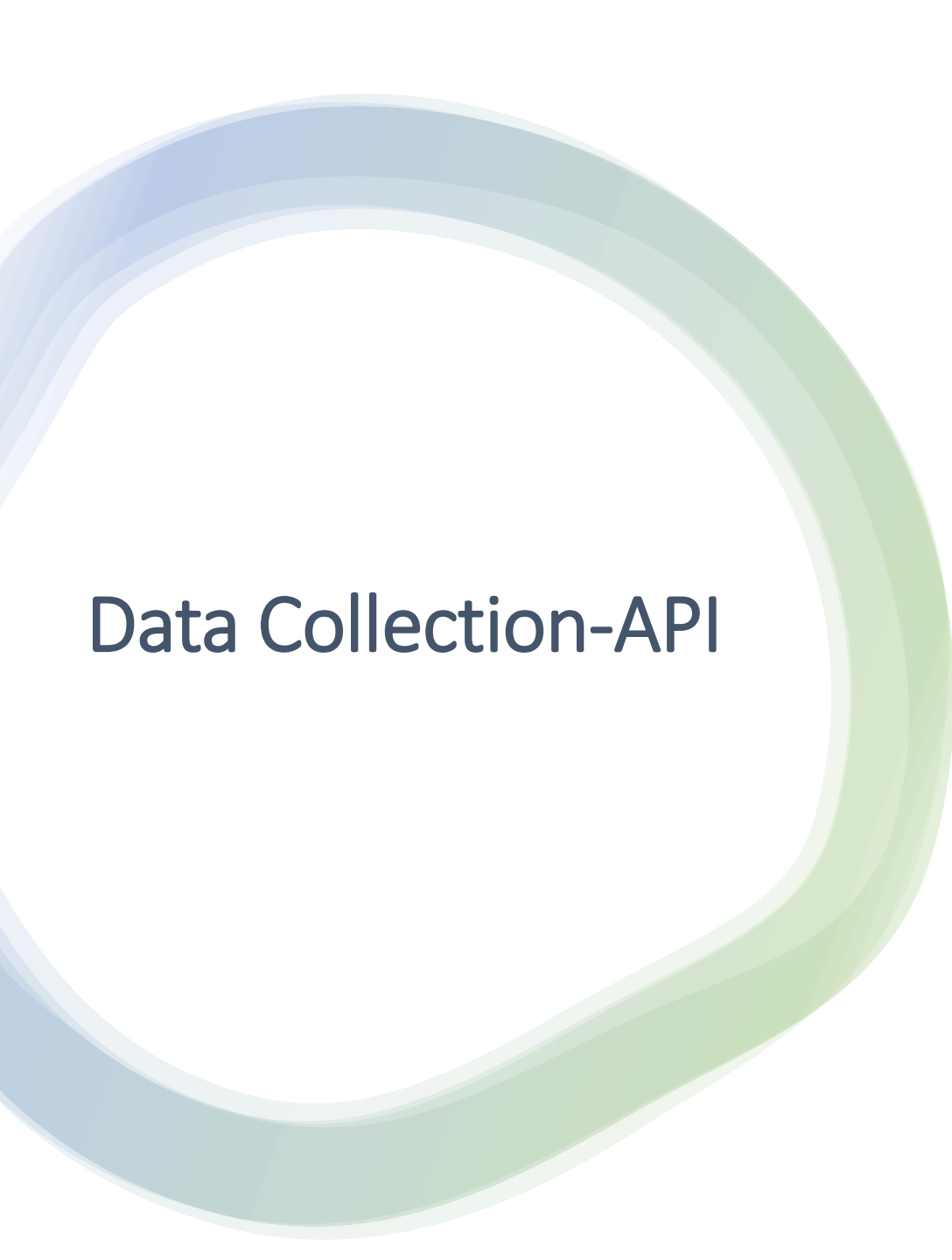


Methodology

- Executive Summary
- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- I used SpaceX REST API to gather data on rocket launches.
- <https://api.spacexdata.com/v4>
- I used webscraping to scrape data from
- <https://en.wikipedia.org/w/index.php?title=List of Falcon 9 and Falcon Heavy launches&oldid=1027686922>



Data Collection-API

- Request Data from SpaceX API
- Decode Response by converting to a JSON file.
- Request Information from SpaceX API.
- Create a dictionary.
- Convert the dictionary to a data frame.
- Replace NAN values and filter data to Falcon9 data.
- View my notebook at:
- <https://github.com/Smermingis1/automatic-potato/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>



Data Collection- Webscrapping

- Request Data from Wikipedia
- Create a BeautifulSoup object.
- Extract Column names from the results.
- Collect the data.
- Create a dictionary from data.
- Create a data frame from the dictionary that was created.
- View my notebook at:
<https://github.com/Smermingis1/automatic-potato/blob/main/jupyter-labs-webscrapping.ipynb>



Data Wrangling

- Eliminate blank and NAN values in the data.
- Convert landing outcomes into a binary column that tells whether a rocket landed or not.
- The data has been prepared for exploratory data analysis.
- The notebook and more details can be found at
- <https://github.com/Smermingis1/automatic-potato/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>



EDA with Data Visualization

Visualizations used on EDA:

1. Flight Number vs. Payload
 2. Flight Number vs. Launch Site
 3. Payload mass(kg) vs. Launch Site
 4. Payload mass(kg) vs. Orbit Type
- You can view my notebook at:
 - <https://github.com/Smermingis1/automatic-potato/blob/main/edadataviz.ipynb>



EDA with SQL

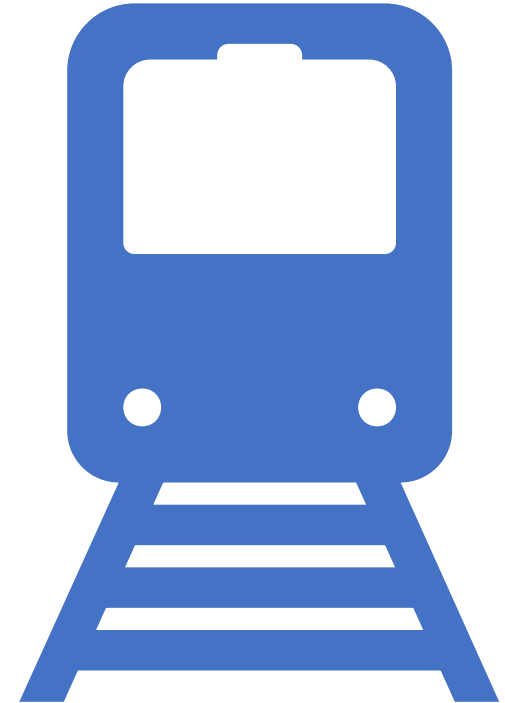
- Things I queried with SQL
- Names of unique launch sites.
- Total payload mass carried by boosters.
- Average payload mass carried by booster version F9 v1.1.
- Count of landing outcomes from June 4, 2010, to March 20, 2017.
- View my notebook at:
- https://github.com/Smermingis1/automatic-potato/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

- Created interactive maps with Folium
- Visualized launch outcomes from different launch sites.
- Used lines and points to measure proximity to:
 - Cities
 - Highways
 - Railways
 - Coastlines

View my notebook data at:

https://github.com/Smermingis1/automatic-potato/blob/main/lab_jupyter_launch_site_location.ipynb





Build a Dashboard with Plotly Dash

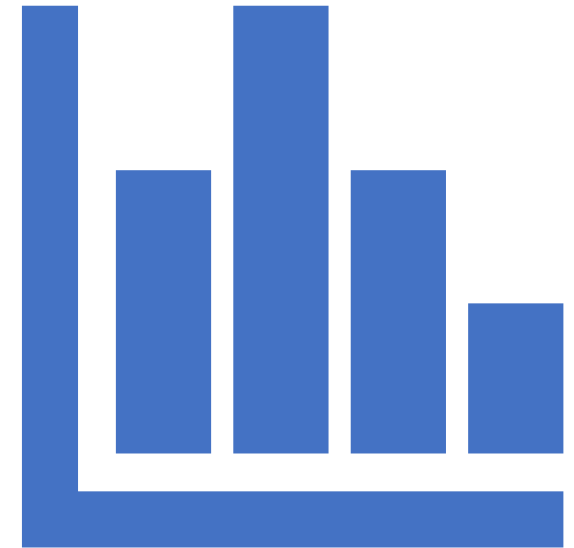
- Built an interactive dashboard with Plotly Dash.
- This displays:
 - A dropdown list of launch sites.
 - A pie chart of successful launches.
 - A scatter chart showing payload mass vs. success rate by booster version.
 - A slider showing the payload mass range.

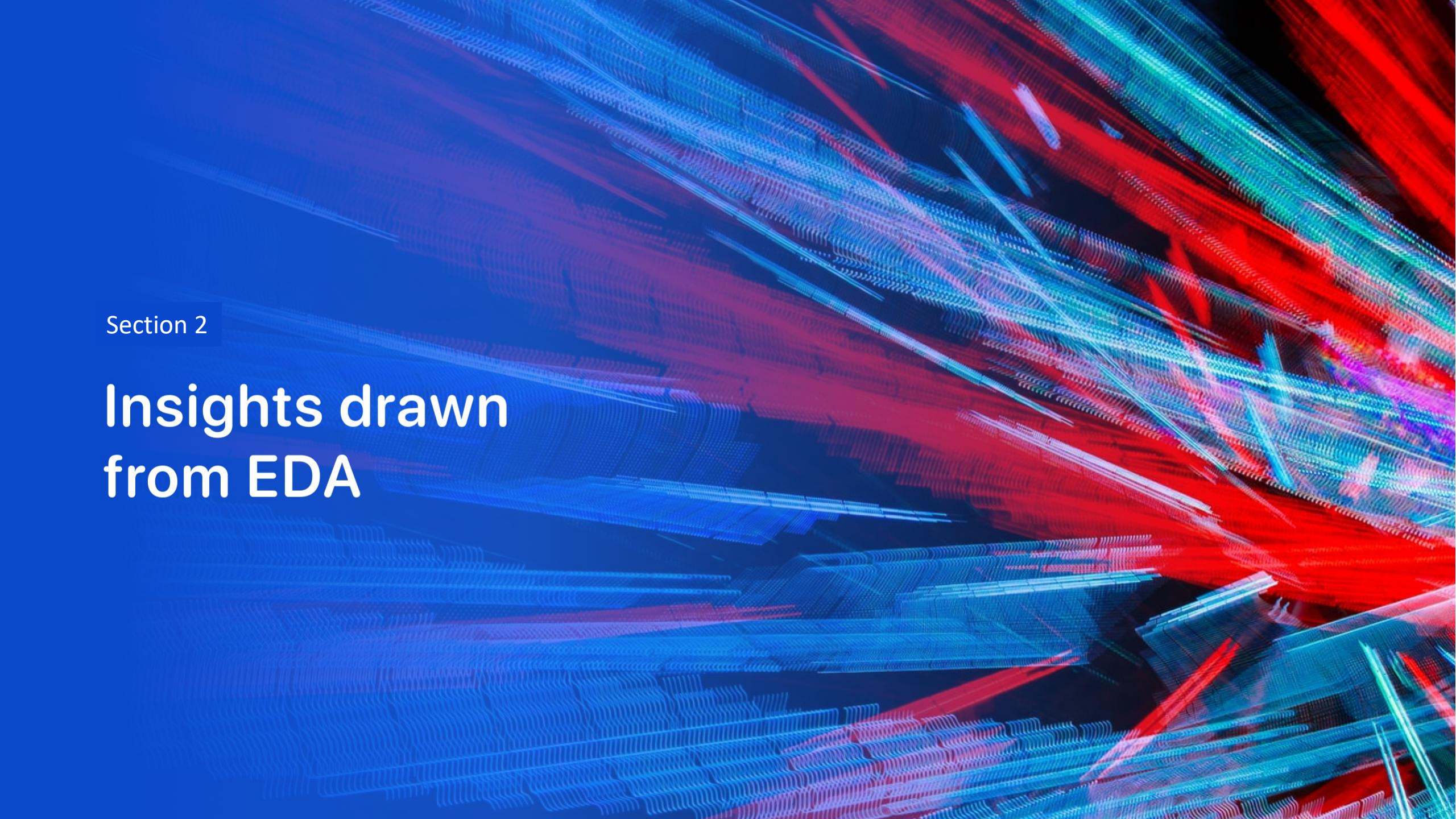
Predictive Analysis (Classification)

- Developed a machine learning model to forecast the success of rocket launches.
- Performed data standardization.
- Divided the dataset into training and testing subsets for model evaluation.
- Trained the model using the prepared data.
- Utilized GridSearchCV to optimize hyperparameters across various algorithms, including:
- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors
- Access the full notebook here:
https://github.com/Smermingis1/automatic-potato/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

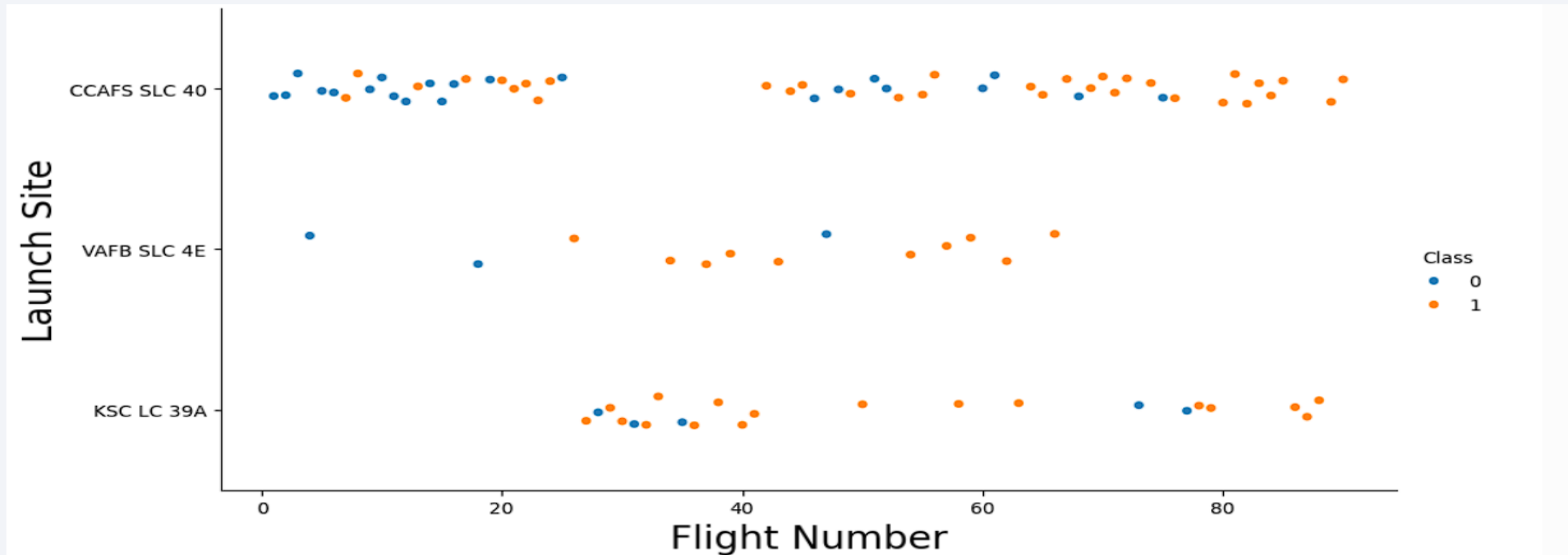


The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

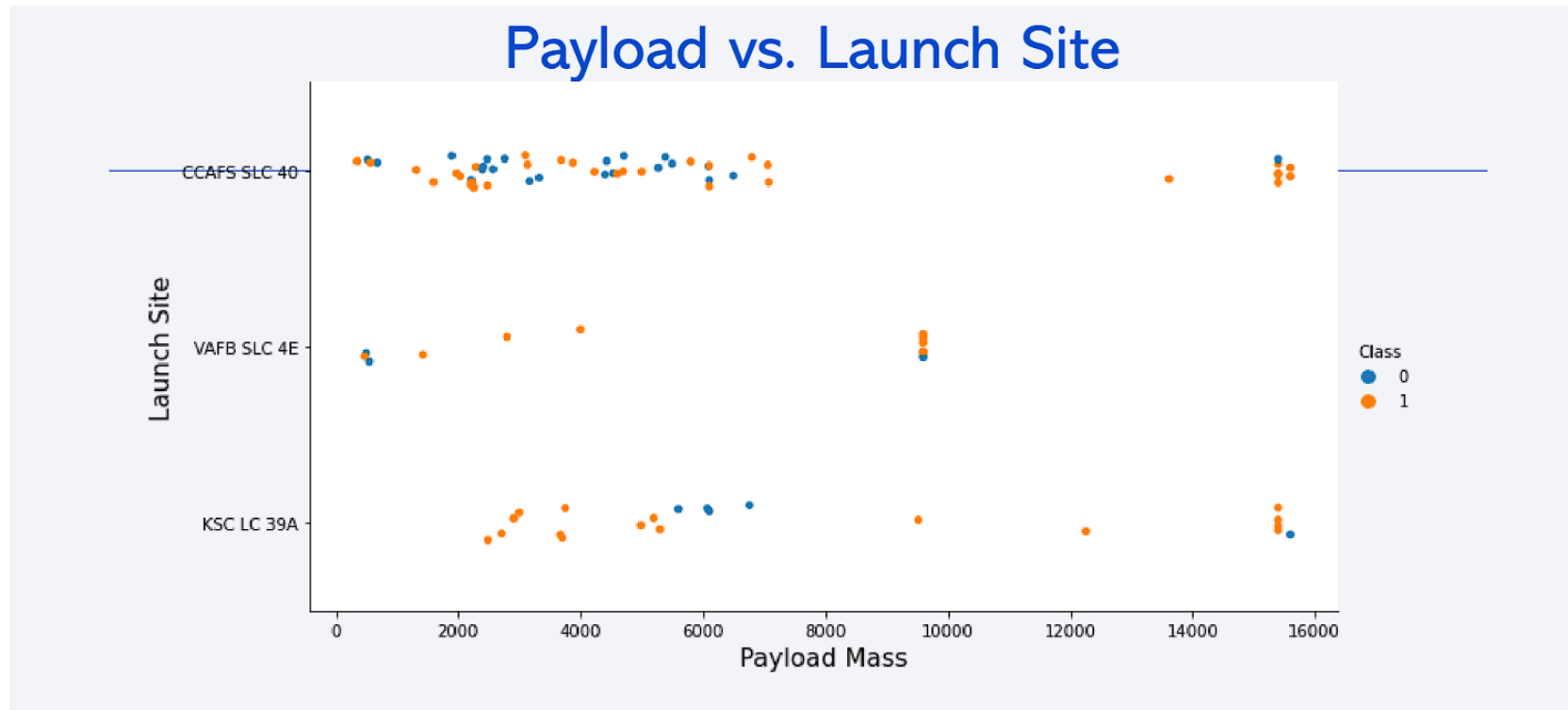
Flight Number vs. Launch Site



- As more flights occur at a given site, the number of successful launches increases.

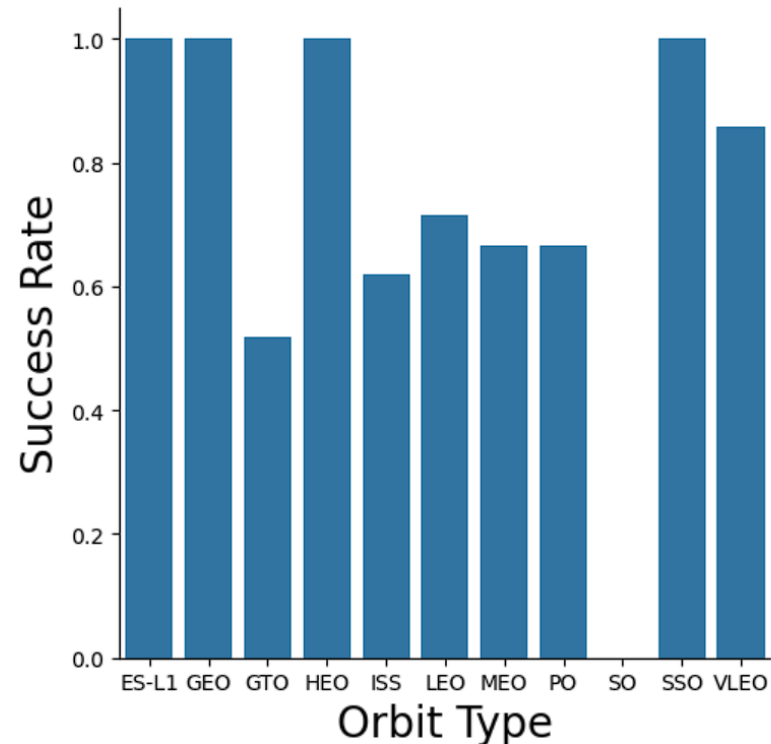
Payload vs. Launch Site

As payload mass increases, the probability of a successful landing also increases.



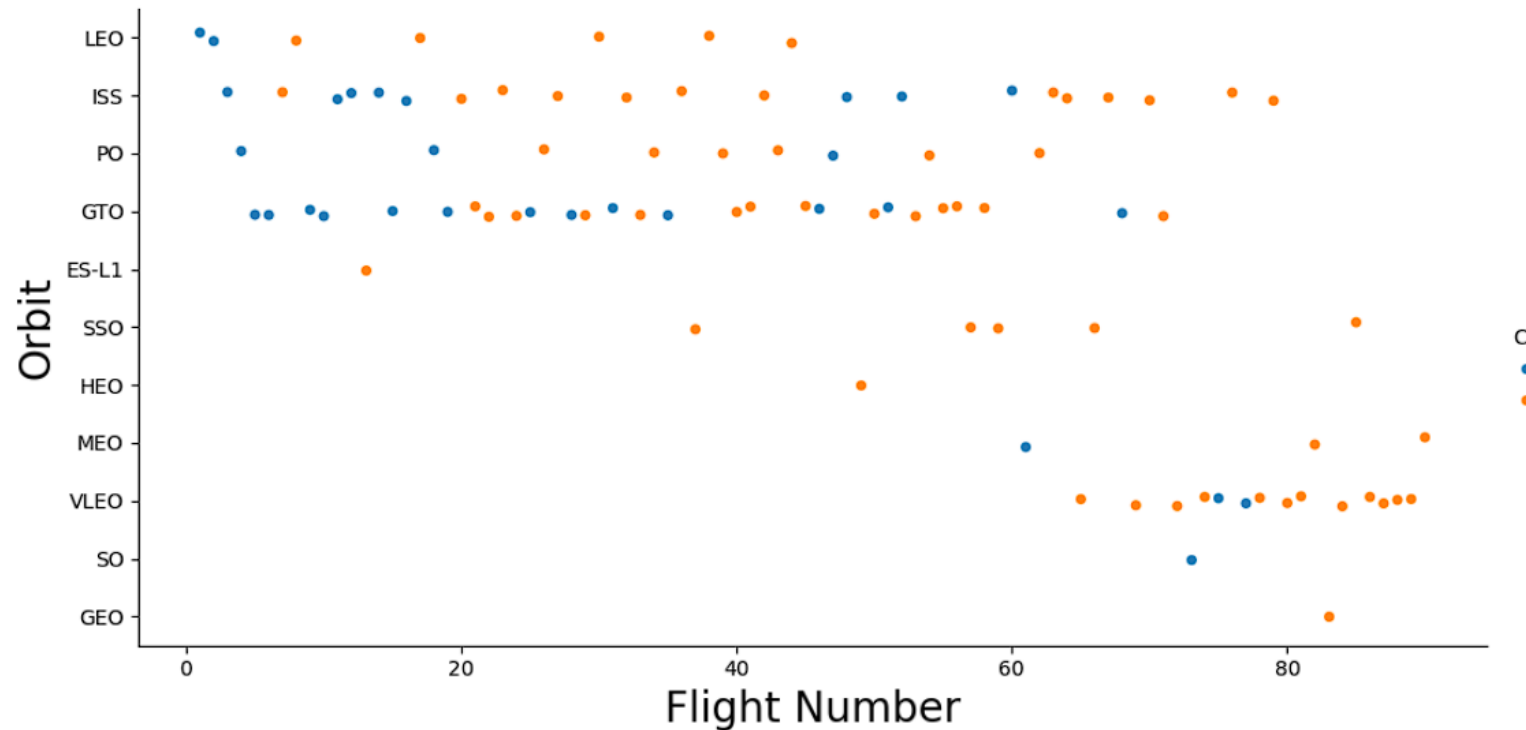
Success Rate vs. Orbit Type

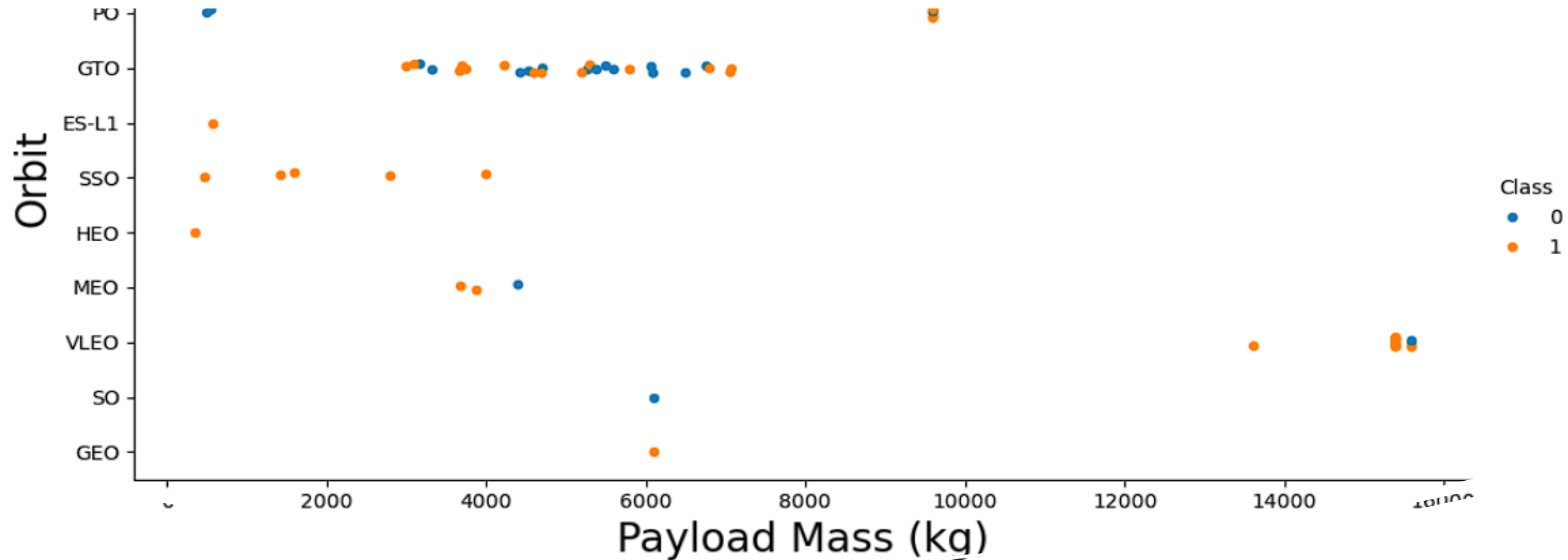
ES-L1, GEO, HEO, SSO, and VLEO have high success rates.



Flight Number vs. Orbit Type

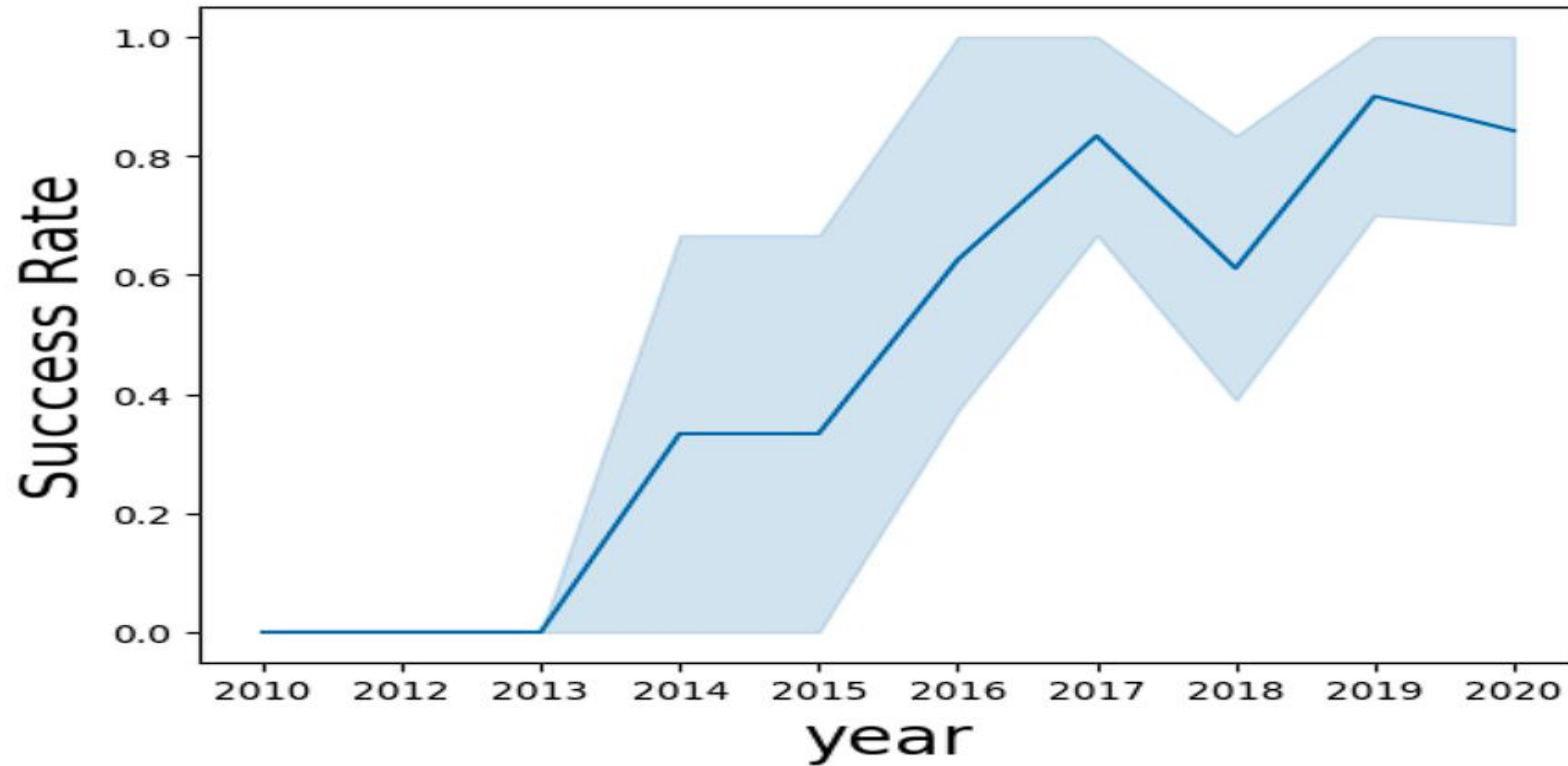
- LEO flights have a success rate tied to the number of flights.
 - In GTO flights, there is no relationship.





Payload vs. Orbit Type

- Heavy payloads are better for LEO, ISS, and PO orbits.
- GTO flights have mixed success.



Launch Success Yearly Trend

Launch success is increasing every year.

Display the names of the unique launch sites in the space mission

```
In [10]: task_1 = '''  
         SELECT DISTINCT LaunchSite  
         FROM SpaceX  
         ...  
         create_pandas_df(task_1, database=conn)
```

```
Out[10]:
```

	launchsite
0	KSC LC-39A
1	CCAFS LC-40
2	CCAFS SLC-40
3	VAFB SLC-4E

All Launch Site Names



Using the SQL command DISTINCT extracted the launch site names.

Launch Site Names Begin with 'CCA'

I extracted 5 records of sites that begin with CCA.

Display 5 records where launch sites begin with the string 'CCA'

```
In [11]: task_2 = '''
        SELECT *
        FROM SpaceX
        WHERE LaunchSite LIKE 'CCA%'
        LIMIT 5
        '''
        create_pandas_df(task_2, database=conn)
```

```
Out[11]:
```

	date	time	boosterversion	launchsite	payload	payloadmasskg	orbit	customer	missionoutcome	landingoutcome
0	2010-04-06	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	2010-08-12	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	2012-05-22	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	2012-08-10	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
4	2013-01-03	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Calculated total payload mass carried by NASA.

In [12]:

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) \
      FROM SPACEXTBL \
      WHERE CUSTOMER = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
Done.
```

Out[12]:

```
SUM(PAYLOAD_MASS__KG_)
```

45596

Average Payload Mass by F9 v1.1

Average payload mass carried by booster version F9 v1.

Display average payload mass carried by booster version F9 v1.1

```
In [13]: %sql SELECT AVG(PAYLOAD_MASS__KG_) \
          FROM SPACEXTBL \
          WHERE BOOSTER_VERSION = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

Done.

```
Out[13]: AVG(PAYLOAD_MASS__KG_)
          2928.4
```

First Successful Ground Landing Date

Dates of the first successful landing outcome on ground pad

In [24]:

```
%sql SELECT MIN(DATE) \  
FROM SPACEXTBL \  
WHERE LANDING_OUTCOME = 'Success (ground pad)'
```

```
* sqlite:///my_data1.db  
Done.
```

Out[24]: MIN(DATE)

2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- Names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

List the names of the boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
L]: %sql SELECT PAYLOAD \
FROM SPACEXTBL \
WHERE LANDING_OUTCOME = 'Success (drone
AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000
```

```
* sqlite:///my_data1.db
Done.
```

```
L]:
```

Payload
JCSAT-14
JCSAT-16
SES-10
SES-11 / EchoStar 105

Total Number of Successful and Failure Mission Outcomes

Total number of successful and failure mission outcomes

List the total number of successful and failure mission outcomes

```
In [26]: %sql SELECT MISSION_OUTCOME, COUNT(*) as total_number \
          FROM SPACEXTBL \
          GROUP BY MISSION_OUTCOME;
```

```
* sqlite:///my_data1.db
Done.
```

Out[26]:

Mission_Outcome	total_number
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Names of the booster which have carried the maximum payload mass.

```
In [28]: %sql SELECT BOOSTER_VERSION \
          FROM SPACEXTBL \
          WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[28]: Booster_Version
```

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

- List of failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

ing_outcomes in drone ship, their booster ve

```
BoosterVersion, LaunchSite, LandingOutcome,
aceX
andingOutcome LIKE 'Failure (drone ship)'
Date BETWEEN '2015-01-01' AND '2015-12-31'

f(task_9, database=conn)
```

launchsite	landingoutcome
CCAFS LC-40	Failure (drone ship)
CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad))

```
In [19]: task_10 = '''
          SELECT LandingOutcome, COUNT(LandingOutcome)
          FROM SpaceX
          WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20'
          GROUP BY LandingOutcome
          ORDER BY COUNT(LandingOutcome) DESC
          '''

          create_pandas_df(task_10, database=conn)
```

```
Out[19]:
```

	landingoutcome	count
0	No attempt	10
1	Success (drone ship)	6
2	Failure (drone ship)	5
3	Success (ground pad)	5
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Precluded (drone ship)	1
7	Failure (parachute)	1

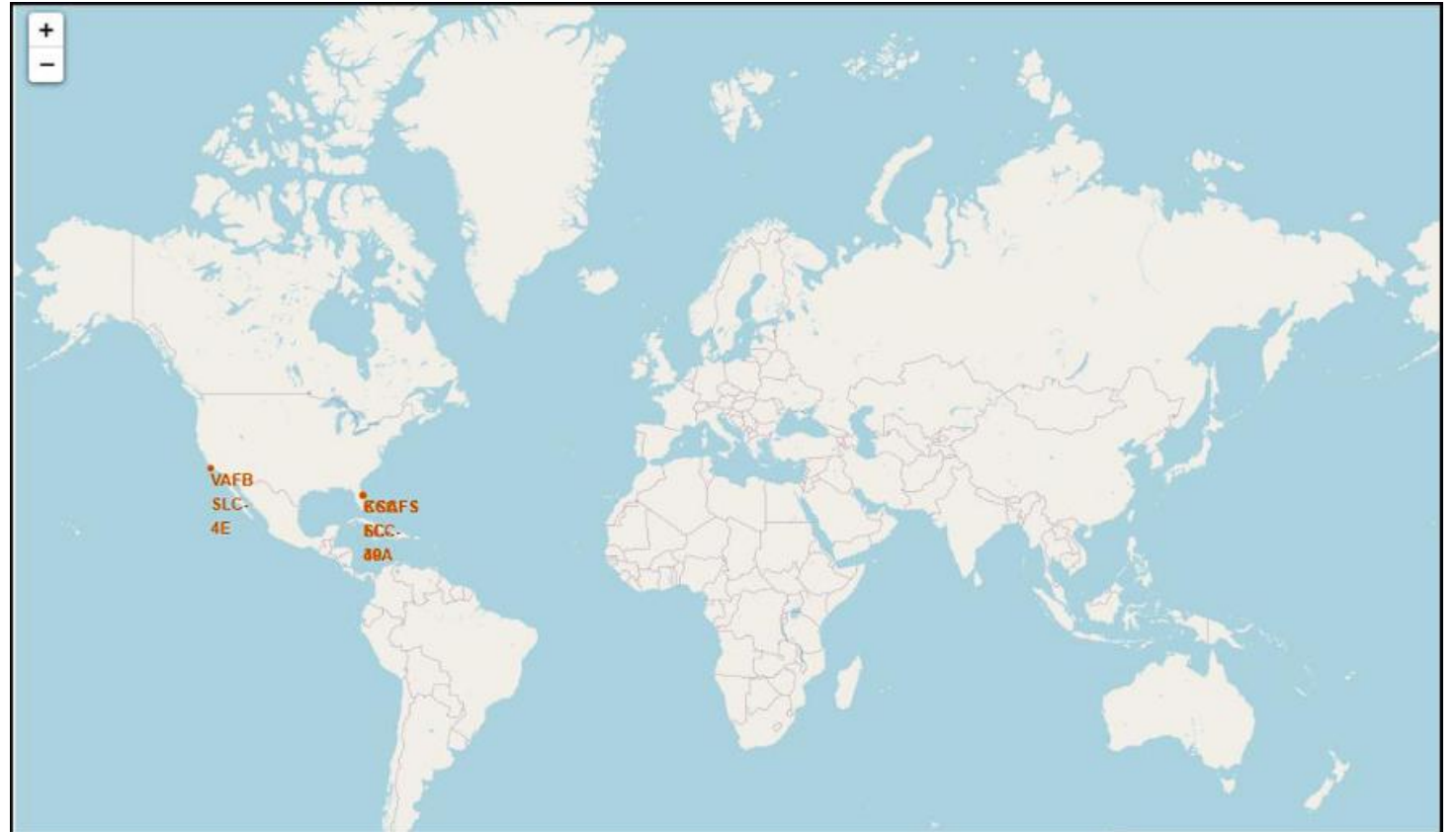
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left shows a clear blue sky.

Section 3

Launch Sites Proximities Analysis

Global SpaceX Operations

- SpaceX has 3 launch sites in Florida and 1 in California.
- They only operate in the United States of America.



Launch Site CCAFS SLC-40

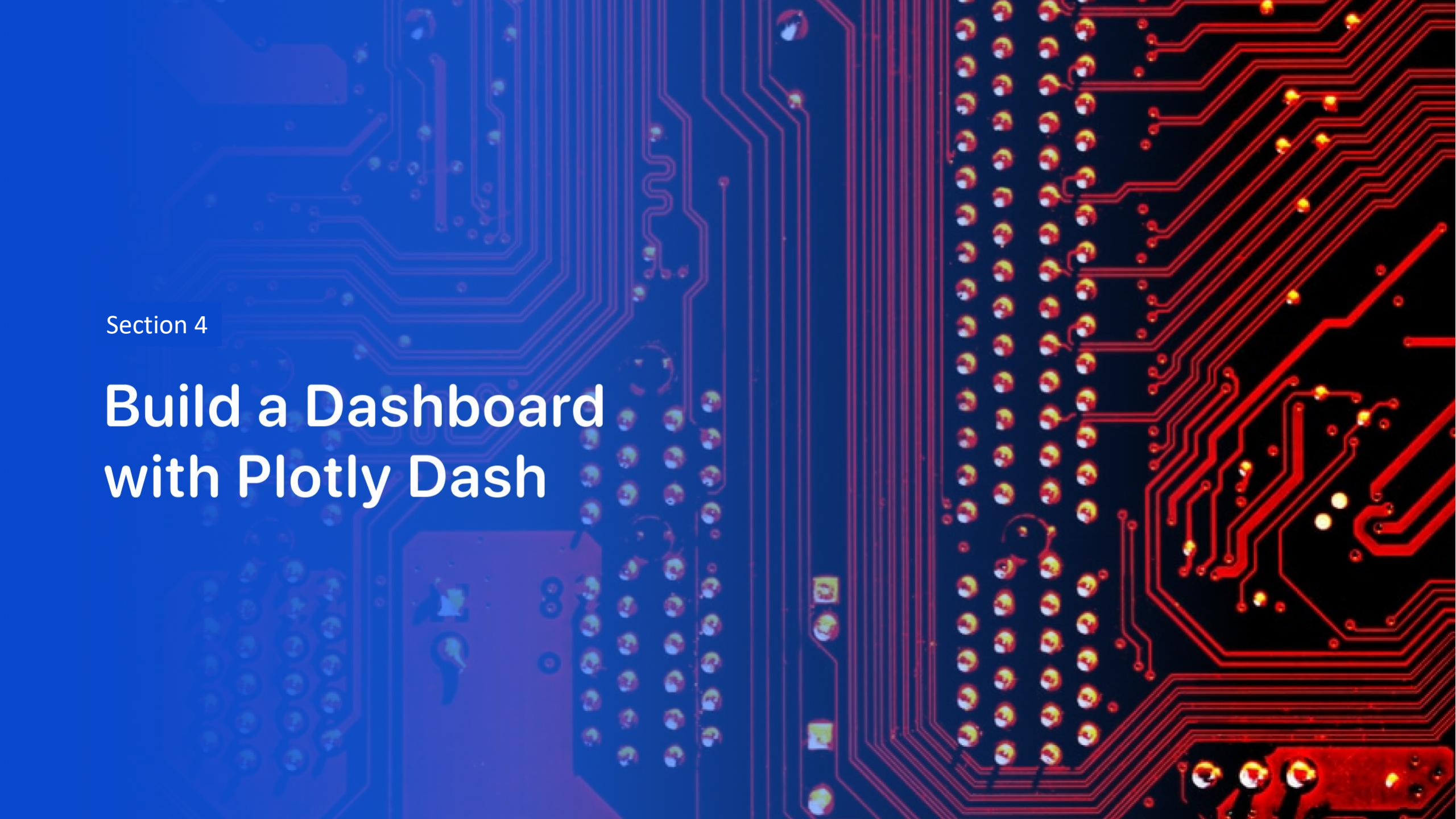
- Red Marker is not a successful launch.
- Green marker is a successful launch.





Distances to Various Features of the Map

- Distance to:
- Nearest Coastline: 0.86Km
- Nearest Railway: 21.96 km
- Nearest City: 23.23 Km
- Nearest Highway: 26.88 Km

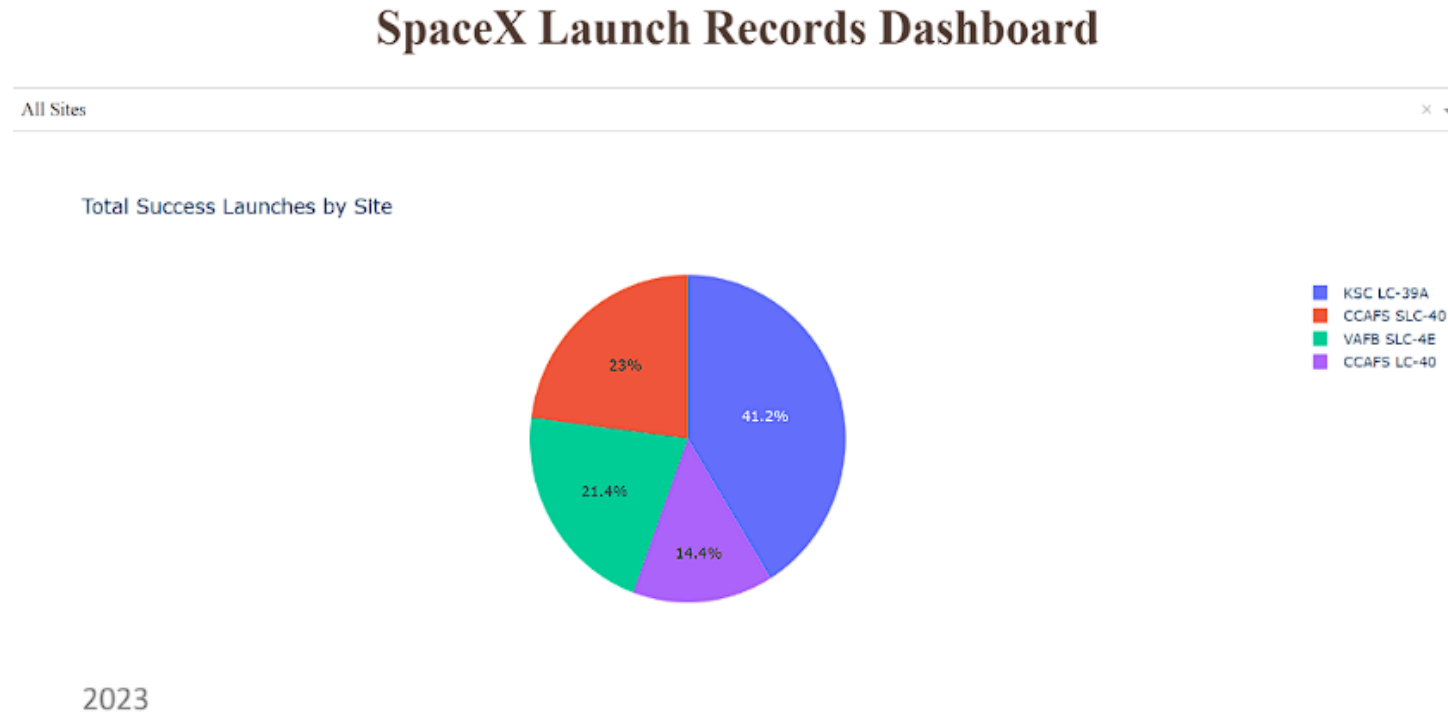


Section 4

Build a Dashboard with Plotly Dash

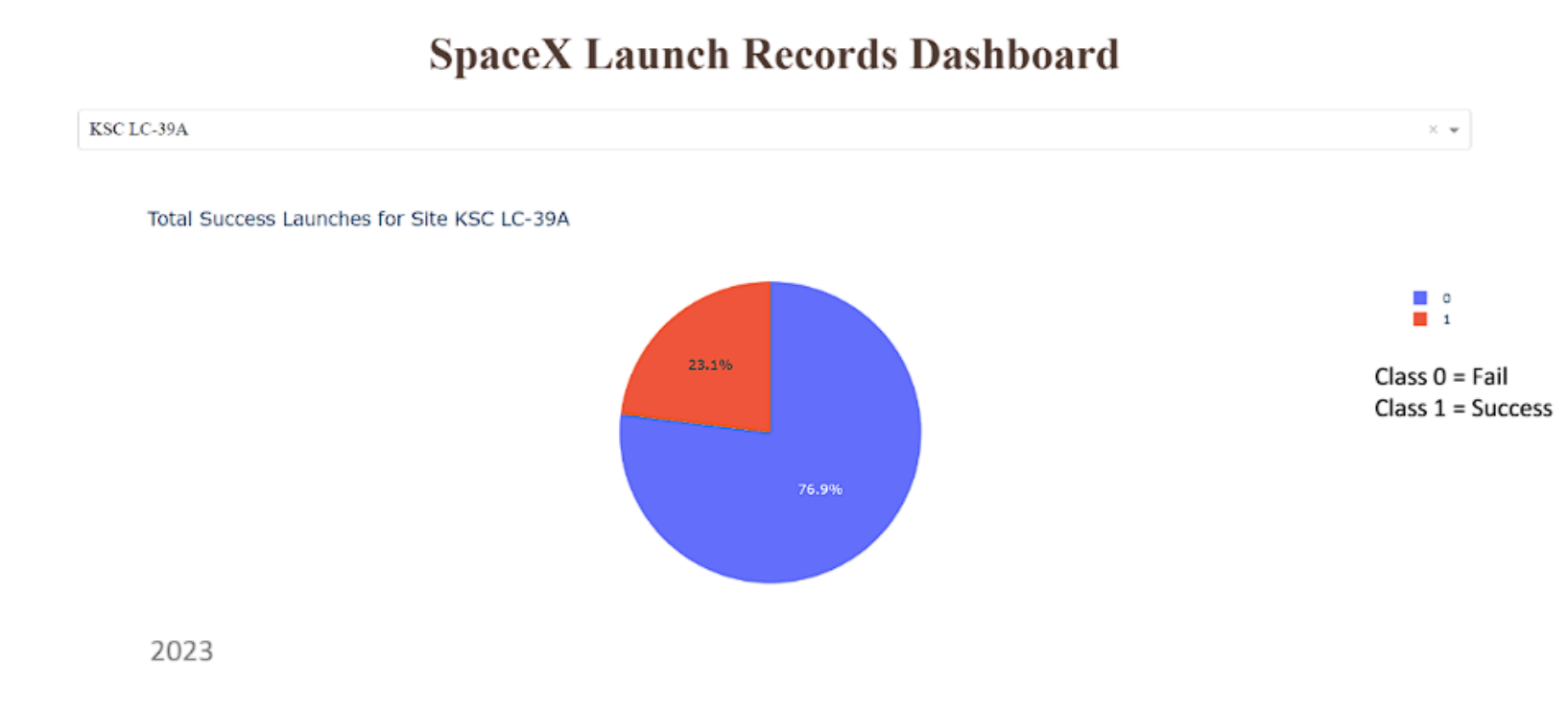
Launch Success by Site PIE chart

This shows the KLC-39A has the most successful launch rate at 41.2%



The Launch Site with the Highest Launch Success Ratio

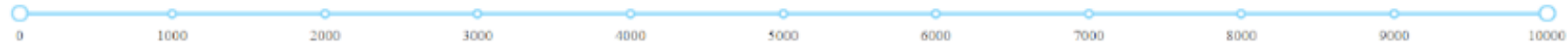
The launch site with the highest launch success ratio is KSC LC-39A.



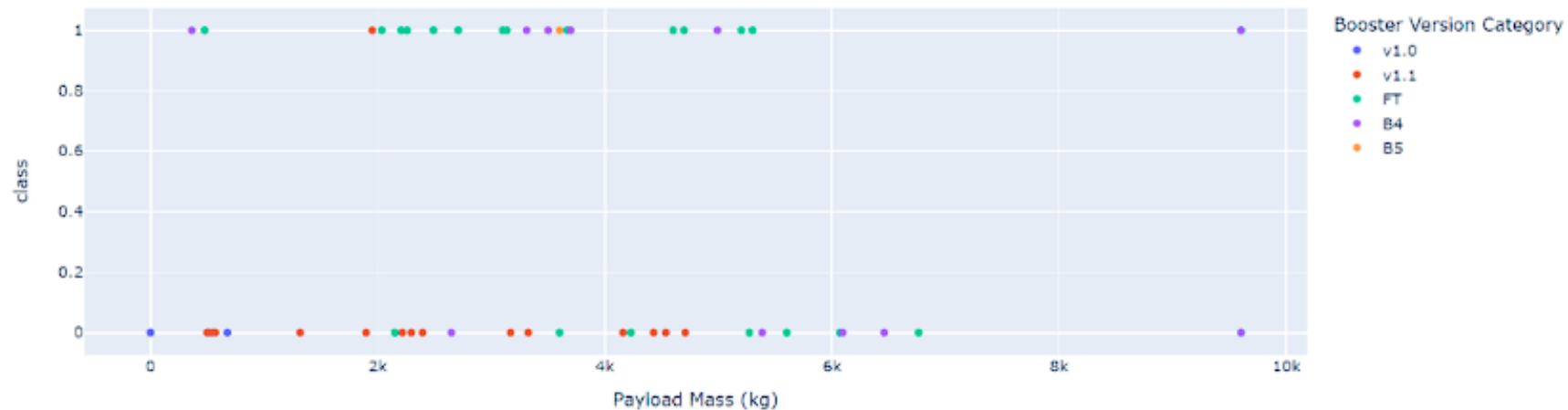
Payload Mass and Success By Booster Version

Payloads between 2000 and 5000 kg have the highest success rate.

Payload range (Kg):



Correlation Between Payload and Success for All Sites

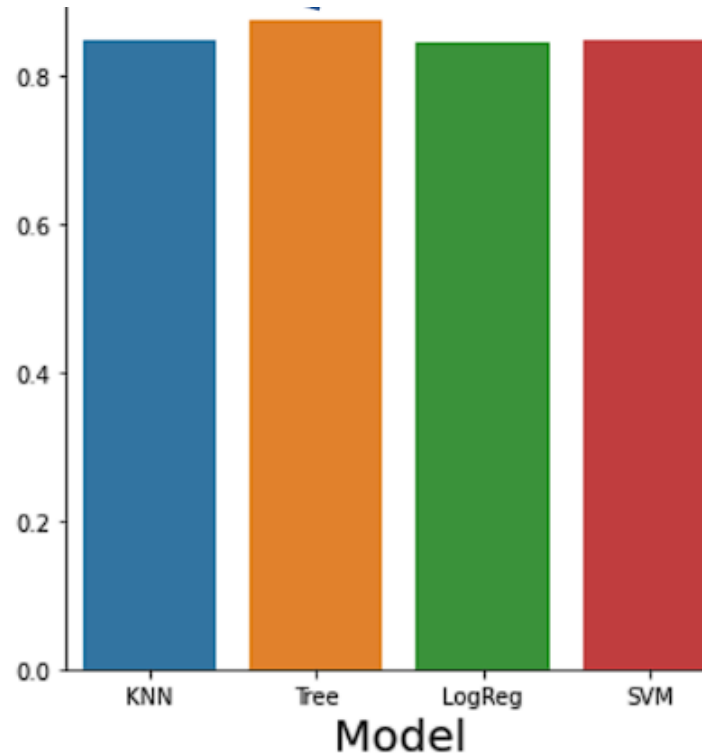


Section 5

Predictive Analysis (Classification)

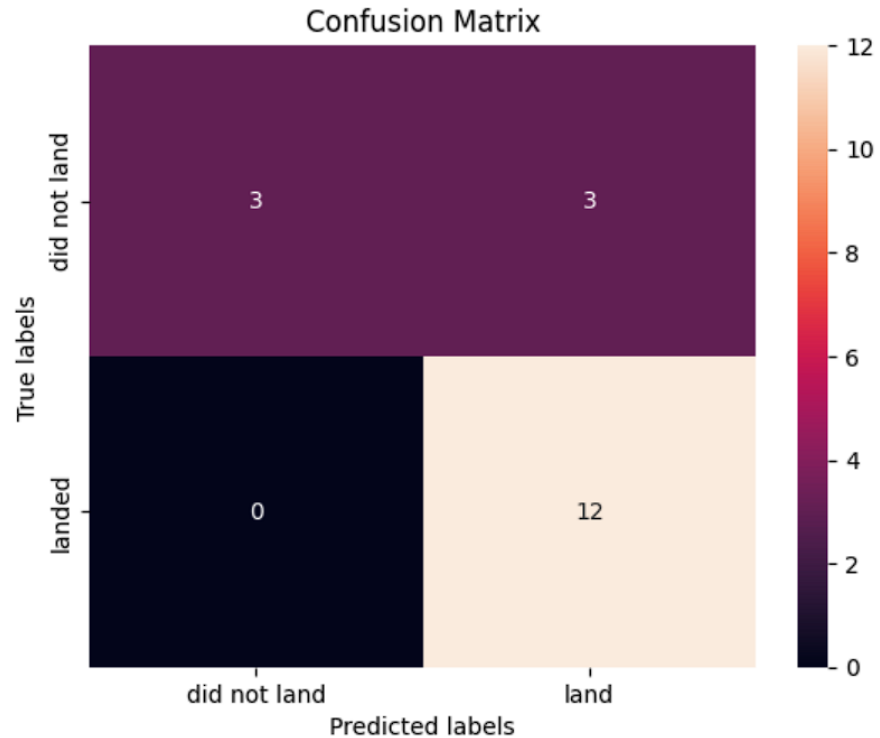
Classification Accuracy

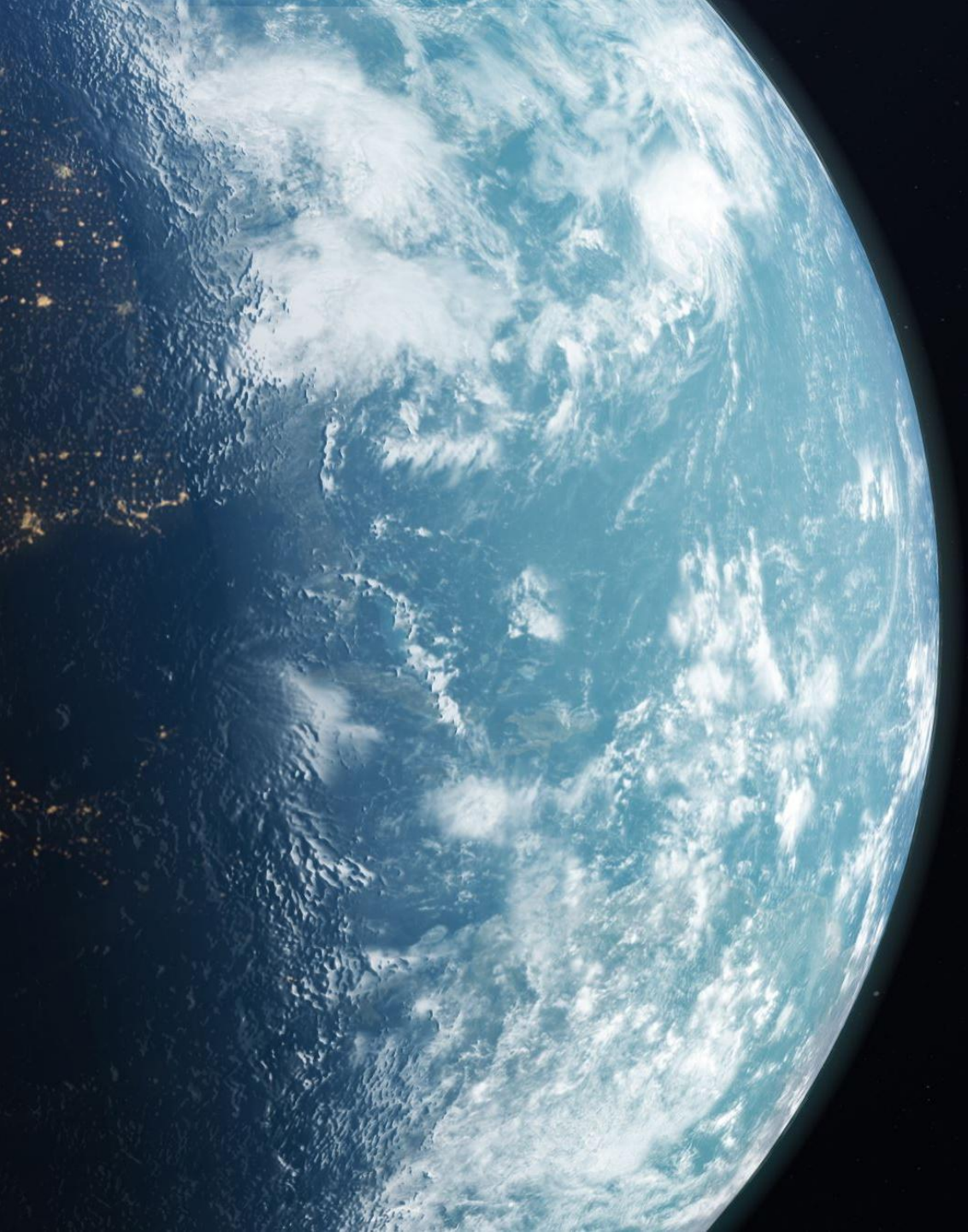
The model with the highest accuracy is the Decision Tree at 0.875.



Confusion Matrix

This is the best confusion matrix that I got in my calculations.





Conclusions

- In conclusion, all launch sites are near a coast and close to the equator.
- Launch success continues to increase over time.
- KSC LC-39A has the highest success rate.
- Orbits ES-L1, GEO, HEO, and SSO have a 100% success rate.
- The higher the payload mass, the higher the success rate.

Thank you!

